

Submersivity: XR-Guided Garbage Detection with an RC Submarine for Peter Street Basin Cleanup

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Abstract—This paper presents an underwater garbage detection and visualization system that combines a remote-controlled submarine, real-time computer vision, basin-scale localization, and an extended reality (XR) head-up display (HUD). A battery-powered submersible based on the open-source ESP-DIVE platform, equipped with a XIAO ESP32-S3 Sense camera, captures underwater imagery and transmits frames for inference using YOLOv8s fine-tuned on a combined terrestrial and underwater debris dataset of 12,177 images. Detected objects are localized within a mapped model of Peter Street Basin (PSB) using structure-from-motion photogrammetry (COLMAP) and ArUco marker registration, then transmitted to Vuzix Smart Swim XR goggles as distance, direction, and class cues for guided cleanup.

The purpose of the system is to make underwater litter survey and recovery more interactive, more intuitive, and more spatially aware. In addition to live detections, the user interface displays a reduced map of PSB with accumulated garbage locations and highlights the nearest target using a compass-style indicator. Preliminary field results demonstrate successful detection of underwater debris under challenging visibility conditions, along with real-time visualization of target direction and range on the XR display.

Index Terms—Underwater computer vision, garbage detection, XR, AR, photogrammetry, COLMAP, ArUco, Peter Street Basin, Vuzix

I. INTRODUCTION

Marine debris detection in underwater environments is challenging due to factors such as light attenuation, scattering, color distortion, suspended particles, and wide appearance variation across debris categories. These difficulties motivate task-specific datasets and robust real-time detection pipelines for underwater robotics [1], [2]. At the same time, current underwater photogrammetry and spatial registration methods make it increasingly practical to build site-specific visual maps that can support navigation, inspection, and intervention [3], [4].

This project addresses those needs through a unified pipeline for underwater cleanup assistance in Peter Street Basin, a rectangular waterfront slip located along Toronto’s central waterfront, constituting the northernmost body of water in all of Lake Ontario. Our system integrates four major components:

- 1) a remote-controlled submarine based on the open-source ESP-DIVE platform for underwater image capture,

- 2) a YOLOv8s garbage detector fine-tuned for single-class debris recognition,
- 3) a basin localization layer combining COLMAP structure-from-motion with ArUco fiducial markers for registering detections in PSB coordinates, and
- 4) an XR HUD deployed on the Vuzix Smart Swim goggles [5] that presents the nearest target’s direction, distance, and class to the user.

The result is a human-in-the-loop cleanup interface in which the sensing platform, the detection model, and the XR display operate together as a single spatial computing system for submerged waste awareness and retrieval.

II. RELATED WORK

A. Underwater garbage detection

Underwater debris detection has emerged as a distinct computer vision problem because image quality is often severely degraded by light attenuation, scattering, turbidity, and color shift. TrashCan introduced a dedicated underwater marine debris dataset with annotations for detection and segmentation, establishing an important benchmark for robot-deployable marine-debris perception [1]. Trash Annotations in Context (TACO) provides a community-contributed terrestrial litter dataset of approximately 1,500 images across fine-grained categories, and has been widely used as a transfer-learning source for waste detection, although it does not directly represent underwater imaging conditions [6]. More recent work further shows that cross-site variation, camera differences, and environmental conditions remain major challenges for robust underwater waste detection, even when modern deep detectors are used [2]. In this context, YOLO-style one-stage detectors remain attractive because they offer a practical speed-accuracy tradeoff for real-time operation [7], [8]. These trends motivate our use of a deployment-oriented detection pipeline that prioritizes robust target discovery in challenging underwater scenes over overly fine-grained waste taxonomy.

B. Structure-from-motion and spatial registration

Structure-from-motion (SfM) photogrammetry has become an effective low-cost approach for constructing detailed spatial models from image collections. COLMAP [9], [10] provides a widely used open-source SfM and multi-view stereo pipeline for automatic feature extraction, matching, and incremental

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sparse reconstruction. Prior work has also demonstrated that SfM photogrammetry can be applied successfully in aquatic environments, including large-scale reef analysis and related marine mapping tasks [3]. At the same time, underwater reconstruction introduces additional geometric challenges due to refraction through flat-port and dome-port housings, motivating recent refractive extensions of standard SfM pipelines [3]. For underwater and close-range applications, marker-based methods using fiducial targets such as ArUco markers [11] provide a practical complement to reconstruction-based methods, particularly when GPS is unavailable and visibility is limited. Prior studies have shown that multiple ArUco markers can support underwater visual navigation and uncertainty-aware pose estimation in challenging conditions [12], [13]. In contrast to work focused primarily on reconstruction or vehicle localization alone, our goal is to register detected debris into a basin-aligned spatial map that can directly support downstream guidance and retrieval.

C. XR guidance for aquatic tasks

Wearable and goggle-mounted head-up systems provide a compelling interface for aquatic tasks because they preserve hands-free operation while maintaining situational awareness in visually constrained environments. Previous works demonstrated the SafeSwim system, which streams overhead drone footage to Vuzix Smart Swim goggles for swimmer safety and situational awareness, showing that Android-based swim goggles can serve as a programmable XR platform for water-based applications [14], [15]. More broadly, aquatic smart-display systems have primarily been explored for training feedback, safety cues, and basic navigation overlays. Our work extends this direction by using the same XR platform not merely as an information display, but as an action-oriented interface that presents distance, direction, and map-based cues derived from underwater debris detection and spatial registration. In this way, the XR layer acts as a bridge between perception and physical cleanup action rather than as a passive visualization module.

III. SYSTEM OVERVIEW

The proposed system forms an end-to-end underwater cleanup assistance pipeline that converts underwater imagery into actionable XR guidance. First, the RC submarine acquires underwater video in Peter Street Basin. Second, the captured frames are transmitted to a host laptop, where a fine-tuned YOLOv8s model detects debris candidates. Third, the accepted detections are localized in basin coordinates and incorporated into a spatial map of PSB. Finally, the system presents the nearest target together with the accumulated mapped detections on XR goggles, providing intuitive distance- and direction-based cues to the user.

Fig. ?? illustrates the data flow across these stages. In this pipeline, the submarine serves as the sensing platform, the host computer performs visual inference and spatial registration, and the XR display translates the resulting map information into real-time user guidance. This end-to-end design allows

raw underwater video to be transformed into spatially meaningful cleanup assistance.

A. RC submarine platform

The sensing platform is based on the ESP-DIVE, an open-source RC submarine. [16] The ESP-DIVE is a compact, battery-powered, dual-ballast submersible with a 3D-printed hull and magnetic drive coupling. The submarine uses a XIAO ESP32S3 Sense module as its onboard computer and camera, which captures video at up to QXGA resolution (2048×1536) through a clear dome port. For stable first-person-view (FPV) streaming during operation, we use SVGA resolution (800×600) at approximately 5 frames per second.

The ESP-DIVE's buoyancy is controlled by dual piston ballast tanks driven by geared motors, allowing the submarine to dive, surface, and maintain neutral buoyancy at a target depth. Steering is provided by a single rear thruster with a rudder. The submarine communicates over Wi-Fi via a tethered antenna buoy connected by a 5-meter coaxial cable, which keeps the RF link above the water surface while the hull operates below. Key operating specifications include a maximum depth of 5 m, a control range of approximately 50 m from the antenna buoy, and a battery life of roughly 30 minutes per charge.

The ESP32S3 family is well suited to this type of compact wireless embedded platform because it combines microcontroller functionality with integrated wireless connectivity and flexible peripheral support [16]. In our system, the submarine is used for underwater image acquisition and streaming, while debris detection is performed on a host laptop. This design choice reflects the computational demands of real-time YOLO inference, which exceed the practical onboard capability of the ESP32S3.

B. Detection model

Our detection module is based on YOLOv8s [8] and is trained as a single-class detector in which all debris categories are merged into a unified “garbage” label. This formulation was adopted after preliminary experiments on the TACO dataset [?] showed that fine-grained category distinctions reduced overall detection performance without providing meaningful benefit for the cleanup task. In particular, training with `single_cls=True` improved mAP50 on the TACO validation split from 0.27 to 0.49.

The final model was fine-tuned on a combined dataset of 12,177 garbage images assembled from multiple sources, including TACO ($\sim 1,500$ terrestrial litter images), the TrashCan underwater debris dataset [1], approximately 1,000 additional underwater garbage images collected by our team, and other underwater garbage images obtained from public repositories. Among the full training set, 5,720 images correspond to underwater scenes. All annotations were converted to YOLO format and unified under the single garbage class. The detector was initialized from COCO-pretrained weights before domain-specific fine-tuning.

Training used a batch size of 8, an input resolution of 640×640 , and standard YOLOv8 data augmentation, including mosaic augmentation, random flipping, HSV perturbation, and random erasing. The final model achieved a precision of 92.1% on the combined validation set. Because underwater imagery is affected by haze, low contrast, and refractive distortion, robust performance depends strongly on domain-specific training rather than direct transfer from terrestrial litter datasets alone [1], [2].

C. Basin map and XR visualization

Photogrammetry and fiducial-marker-based registration are used to construct a spatial model of Peter Street Basin (PSB) and to localize debris detections within a basin-aligned coordinate frame. The submarine's video stream is transmitted to a host laptop, where the detection model performs visual inference and updates a spatial detection map in PSB coordinates. The resulting localized targets are then forwarded to the XR interface, which presents both a reduced PSB map and a compass-like directional cue indicating the nearest debris item.

IV. DETECTION AND LOCALIZATION PIPELINE

A. Image acquisition and inference

Let I_t denote the underwater frame acquired at time t . The detector returns a set of garbage hypotheses

$$\mathcal{D}_t = \{(b_i, s_i)\}_{i=1}^{N_t}, \quad (1)$$

where b_i is a bounding box in image coordinates, s_i is the corresponding confidence score, and N_t is the number of detections at time t . We retain detections above a confidence threshold τ and forward them to the localization module.

For the preliminary batch survey used in our field analysis, frames were extracted from GoPro video recorded during a perimeter walk of Peter Street Basin using a dual-camera rig. The above-water camera captured basin walls and ArUco markers for spatial registration, while the underwater camera was mounted with an approximately 45° downward offset toward the submerged wall. The underwater camera was housed in a dome port to mitigate refraction-related image distortion.

B. Structure-from-motion reconstruction

To estimate camera poses along the survey trajectory, we applied COLMAP-based structure-from-motion reconstruction to the above-water GoPro image sequence. The reconstruction pipeline consisted of three stages:

a) *Feature extraction*: SIFT keypoints and descriptors were extracted from each frame using COLMAP's GPU-accelerated feature extractor. The camera model was set to `OPENCV_FISHEYE` to account for the GoPro's wide-angle lens. COLMAP jointly optimized the camera intrinsics during reconstruction, yielding focal lengths of $f_x = 2440.5$, $f_y = 2445.0$ and four fisheye distortion coefficients.

b) *Feature matching*: Sequential matching was used instead of exhaustive matching, exploiting the high overlap between consecutive frames in the perimeter survey. This reduced the number of candidate image pairs from approximately 338,000 to a manageable set of neighboring pairs.

c) *Sparse reconstruction*: The incremental mapper registered 796 of 823 images into a single consistent sparse reconstruction containing approximately 415,000 3D points, with a mean reprojection error of approximately 1 pixel. The remaining 27 images that failed to register were discarded. For each registered frame, COLMAP produced a camera pose $(\mathbf{R}_t, \mathbf{t}_t)$ in a shared world coordinate frame, from which the camera position was recovered as $\mathbf{p}_t = -\mathbf{R}_t^T \mathbf{t}_t$.

The resulting camera trajectory traces the perimeter of Peter Street Basin and provides the spatial backbone for subsequent detection registration. Because the survey followed a slow perimeter path beginning at the southeast corner of the basin, the temporal ordering of frames closely corresponds to spatial progression around PSB.

C. ArUco marker registration

To anchor the COLMAP reconstruction in a real-world basin reference frame, ArUco fiducial markers from the `ARUCO_4X4_50` dictionary (19.5 cm side length) were placed at known locations around the basin walls. The markers were detected in the above-water image sequence using OpenCV's ArUco module, and their known 3D positions in the basin reference frame served as ground-control points for transforming the COLMAP coordinate system into a basin-aligned coordinate frame.

Marker locations were obtained by labeling their positions on a pre-existing Blender 3D model of the basin, which provided a more precise geometric reference than manual field measurement. A rigid transformation was then estimated from the corresponding marker positions in the COLMAP and basin frames. An additional rotation was applied to align the basin's long axis and to account for the approximately 25° offset between the basin reference frame and geographic north.

D. Basin-coordinate registration

Let the submarine (or camera) pose in basin coordinates at time t be

$$\mathbf{p}_t = [x_t, y_t, z_t]^T, \quad \psi_t \in \mathbb{R}, \quad (2)$$

where (x_t, y_t) are planar basin coordinates, z_t is depth, and ψ_t is heading. For each accepted detection, we estimate a relative bearing θ_i and range r_i from the image geometry and current calibration. The corresponding basin-frame point is modeled as

$$\mathbf{g}_i = \begin{bmatrix} x_t + r_i \cos(\psi_t + \theta_i) \\ y_t + r_i \sin(\psi_t + \theta_i) \\ z_t + \Delta z_i \end{bmatrix}, \quad (3)$$

where Δz_i captures the target's relative vertical offset. Each registered garbage hypothesis is then appended to the basin map.

For the dual-camera rig used during the batch survey, the underwater camera’s pose was derived from the above-water camera’s ArUco-based or COLMAP-based pose using a fixed rig transform. The underwater camera was offset approximately 0.3 m (1 foot) at a 45° angle to the left and downward relative to the above-water camera, as determined by the handheld rig geometry.

E. Nearest-target cueing

If the current set of mapped targets is $\mathcal{G} = \{\mathbf{g}_k\}$, the XR system selects the next target by nearest-neighbor distance:

$$k^* = \arg \min_k \|\mathbf{g}_k - \mathbf{p}_t\|_2. \quad (4)$$

The displayed target distance is

$$d^* = \|\mathbf{g}_{k^*} - \mathbf{p}_t\|_2, \quad (5)$$

and the relative planar bearing shown to the user is

$$\phi^* = \text{atan2}(y_{k^*} - y_t, x_{k^*} - x_t) - \psi_t. \quad (6)$$

This allows the XR HUD to point directly toward the next collection candidate while simultaneously displaying a global basin overview.

V. XR INTERFACE

The XR interface is implemented as a custom Android application deployed on Vuzix Smart Swim goggles. Smart Swim is an Android-based wearable head-up display with integrated wireless connectivity and a monocular full-color display, making it suitable as a programmable XR platform for aquatic use [5]. Although the device was originally designed for lap-swimming metrics, its Android operating system and wireless communication capabilities enable real-time visualization of external perception outputs, as also demonstrated in prior aquatic XR systems such as SafeSwim [15].

The display presents three core elements:

- 1) a compass-like directional pointer to the nearest target,
- 2) a range readout in meters, and
- 3) a reduced PSB map showing the current submarine position and previously registered detections.

The interface is intentionally minimal, translating the perception pipeline into concise action-oriented guidance rather than exposing raw detection outputs. Specifically, the user is shown the nearest debris target, its distance, and the direction in which to move. This design is particularly appropriate for underwater operation, where visibility is limited and continuously updated but compact guidance is preferable to dense visual overlays.

Because the display is monocular, it preserves part of the user’s natural field of view, which is beneficial for safety and spatial awareness during swimming. The application receives basin-coordinate target updates from the host laptop over Wi-Fi and renders the directional and distance cues at an update rate of approximately 5–10 Hz, which is sufficient for real-time user guidance in the intended deployment setting.

TABLE I: Detection model training progression.

Configuration	Images	mAP50	Precision
YOLOv8n, TACO multi-class	1,500	0.27	0.37
YOLOv8s, TACO single-class	1,500	0.49	0.71
YOLOv8s, combined single-class	12,177	—	0.921

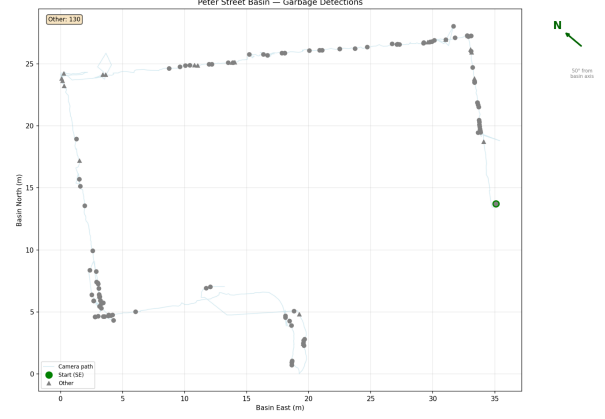


Fig. 1: Representative Peter Street Basin survey path with accumulated garbage detections registered in basin coordinates. Camera trajectory derived from COLMAP sparse reconstruction of 796 registered frames.

VI. PRELIMINARY RESULTS

A. Detection model performance

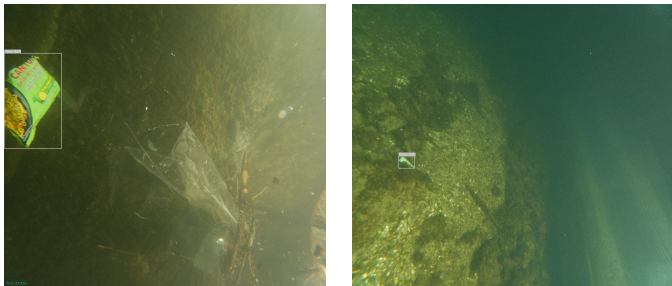
Table I summarizes the training progression of the YOLOv8s model. The initial baseline trained on TACO alone with multi-class labels achieved modest performance (mAP50 = 0.27), primarily because the model struggled to distinguish fine-grained categories (bottle vs. can vs. cup), which are irrelevant to the cleanup task. In order to improve this, we switched to single-class training and scaled to the larger combined dataset with underwater images, which yielded much better results.

B. Basin mapping results

Fig. 1 shows a representative survey trajectory and the accumulation of mapped detections in Peter Street Basin coordinates. The COLMAP sparse reconstruction registered 796 of 823 above-water frames into a consistent model, from which camera positions were extracted and transformed into basin coordinates using the ArUco anchor points. In the plotted run, the system recorded 130 detector hits in the *Other* class. Because these are frame-level detections along a continuous trajectory, they should be interpreted as detection events rather than de-duplicated physical objects.

C. Qualitative underwater detections

Fig. 2 shows two representative underwater detections. The first example captures packaging-like debris under turbid lighting, while the second shows a smaller distant object detected



(a) Packaging/plastic debris. (b) Small distant detection.

Fig. 2: Example underwater garbage detections captured during PSB deployment.



Fig. 3: XR head-up display on Vuzix Smart Swim goggles showing the nearest target class, range, heading cue, and a reduced PSB map.

near the basin floor. These examples illustrate the diversity of appearance the model must handle, including partial occlusion, low contrast, and significant color shift due to light attenuation and scattering in the basin water.

D. HUD output

Fig. 3 shows the output on the XR interface (Vuzix Smart Swim display) during operation. The display reports the target class as `Other`, indicates a range of 4.1 m, and uses a compass arrow to direct the user toward the selected target while showing a miniaturized PSB map with registered detections. In addition, the display shows

VII. DISCUSSION

The results suggest that underwater cleanup assistance benefits from combining perception and guidance rather than treating object detection as an isolated backend task. The detector alone can identify likely debris, but the addition of basin registration and XR cueing turns those detections into actionable spatial instructions.

The choice to use a single-class detector proved important in practice. Early multi-class experiments on the TACO dataset showed that fine-grained classification hurt overall detection accuracy without providing useful information for cleanup — a swimmer collecting debris does not need to know whether an item is a bottle or a can before picking it up. Collapsing all categories into a single “garbage” class nearly doubled mAP50 and substantially reduced false negatives.

The COLMAP-based localization pipeline offers an offline alternative to real-time odometry. For the preliminary survey, batch-processing the above-water GoPro footage through SfM reconstruction yielded accurate camera trajectories without requiring an IMU or magnetometer on the submarine. However, this approach is only suitable for post-hoc analysis; the live system relies on the submarine’s heading estimates and simplified dead reckoning for real-time position updates, which introduces drift over longer trajectories.

Several additional limitations remain. First, repeated observations of the same debris inflate raw detection counts unless track association or spatial clustering is applied. Second, underwater image degradation still causes false negatives for low-contrast or heavily occluded items. Third, the GoPro dome port used for the underwater camera only partially corrects for refraction, and COLMAP’s intrinsic calibration was performed for the above-water camera only. Fourth, the fixed rig transform between above-water and underwater cameras (estimated at 45° , 0.3 m offset) was measured roughly and introduces localization error in the underwater detections. Fifth, the ESP-DIVE’s maximum depth of 5 m and 30-minute battery life constrain the survey area and duration for deeper or larger water bodies.

Future work will focus on four directions:

- 1) multi-frame association and DBSCAN spatial clustering for de-duplicating mapped objects into confirmed physical debris locations,
- 2) underwater image enhancement or color correction prior to detection to compensate for turbidity and color shift,
- 3) closed-loop collection behaviors in which the submarine not only detects debris but also assists in retrieval planning by prioritizing dense clusters, and
- 4) on-device inference using a TFLite-exported model running directly on the Vuzix Smart Swim’s Android processor, which would eliminate the laptop from the pipeline entirely for simple detection tasks.

VIII. CONCLUSION

This paper presented a garbage detection submarine system that integrates underwater computer vision, basin-scale spatial mapping, and XR visualization for Peter Street Basin cleanup. A compact RC submarine based on the open-source ESP-DIVE platform acquires underwater imagery via a XIAO ESP32S3 Sense camera. A YOLOv8s detector, trained as a single-class model on 12,177 terrestrial and underwater garbage images, identifies marine debris with 92.1% precision. COLMAP structure-from-motion reconstruction and ArUco fiducial markers localize detections in basin coordinates, and Vuzix Smart Swim XR goggles present the nearest target’s direction and distance to the user through a custom Android application. Together, these components form a practical human-in-the-loop pipeline for making submerged cleanup more spatially aware, interactive, and operationally efficient.

IX. LIMITATION AND FUTURE WORK

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REFERENCES

- [1] J. Hong, M. Fulton, and J. Sattar, "Trashcan: A semantically-segmented dataset towards visual detection of marine debris," *arXiv preprint arXiv:2007.08097*, 2020.
- [2] J. S. Walia and K. Seemakurthy, "Optimized custom dataset for efficient detection of underwater trash," in *Towards Autonomous Robotic Systems*, ser. Lecture Notes in Computer Science. Cham: Springer Nature Switzerland, 2023, pp. 292–303.
- [3] D. T. I. Bayley and A. O. M. Mogg, "A protocol for the large-scale analysis of reefs using structure from motion photogrammetry," *Methods in Ecology and Evolution*, vol. 11, no. 11, pp. 1410–1420, 2020.
- [4] J. Wittmann, S. Chatterjee, and T. Sure, "Robust marker detection and identification using deep learning in underwater images for close range photogrammetry," *ISPRS Open Journal of Photogrammetry and Remote Sensing*, vol. 13, p. 100072, 2024.
- [5] Vuzix Corporation, "Vuzix' award-winning smart swim™ smart glasses now shipping," <https://ir.vuzix.com/news-events/press-releases/detail/1752/vuzix-award-winning-smart-swim-smart-glasses-now>, 2020.
- [6] P. F. Proença and P. Simões, "Taco: Trash annotations in context for litter detection," *arXiv preprint arXiv:2003.06975*, 2020.
- [7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 779–788.
- [8] G. Jocher, J. Qiu, and A. Chaurasia, "Ultralytics YOLO," 2026.
- [9] J. L. Schonberger and J.-M. Frahm, "Structure-from-motion revisited," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 4104–4113.
- [10] J. L. Schönberger, E. Zheng, J.-M. Frahm, and M. Pollefeys, "Pixel-wise view selection for unstructured multi-view stereo," in *European conference on computer vision*. Springer, 2016, pp. 501–518.
- [11] S. Garrido-Jurado, R. Muñoz-Salinas, F. J. Madrid-Cuevas, and M. J. Marín-Jiménez, "Automatic generation and detection of highly reliable fiducial markers under occlusion," *Pattern Recognition*, vol. 47, no. 6, pp. 2280–2292, 2014.
- [12] Z. Xu, M. Haroutunian, A. J. Murphy, J. Neasham, and R. Norman, "An underwater visual navigation method based on multiple aruco markers," *Journal of Marine Science and Engineering*, vol. 9, no. 12, p. 1432, 2021.
- [13] P. Risholm, P. Ø. Ivarsen, K. H. Haugholt, and A. Mohammed, "Underwater marker-based pose-estimation with associated uncertainty," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 3713–3721.
- [14] M. She, F. Seegräber, D. Nakath, and K. Köser, "Refractive colmap: refractive structure-from-motion revisited," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 12 816–12 823.
- [15] S. Mann, C. Mann, F. Sadrzadeh-Afsharazar, Z. Lu, J. Saini, and J. Lian, "Safeswim: Overhead vision to help people see better," in *2020 IEEE Congreso Bienal de Argentina (ARGENCON)*. IEEE, 2020, pp. 1–8.
- [16] Espressif Systems, "ESP32 Series Datasheet," https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf, 2025, version 5.2.